T E C M I L E N I O

Sistemas y tecnologías de administración del conocimiento

Document Recommender System

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1. Description of the Approach

The recommender helps people at a Toyota plant find related knowledge: when someone opens a policy, a ticket or types a question, the system returns the **Top-5 most similar documents**. It uses **Path A (TF-IDF + cosine similarity)** because it is simple, fast and transparent: every score can be explained by the words two documents share.

1.1. Repository

The corpus (*corpus.csv*) holds **40 simulated internal documents** with the columns id, title, text and category: Policy/Process (11), Support/Incidents (11), Quality (7), Training (7) and Supply Chain (4). They include policies, IT tickets, incident reports, meeting minutes, FAQs and training modules.

1.2. Pipeline

- **Cleaning:** lowercase, removal of special characters and English stopwords.
- **Vectorization:** *TfidfVectorizer* with unigrams and bigrams over title + text, producing a 40 × 1,722 matrix.
- **Similarity:** cosine similarity between documents, or between the query and every document.
- **Ranking:** Top-5 by score, excluding the seed document. Both modalities are available: document ID and free text.

1.3. Results of the Tests

Test	Input	Top-3 results (score)	Coherent
1. Policy/Process	DOC001 Andon cord response policy	DOC028 (0.188), DOC002 (0.098), DOC019 (0.079)	Yes. First result is the andon training module.
2. Support/Incidents	DOC011 Ticket: MES terminal frozen	DOC032 (0.286), DOC012 (0.156), DOC019 (0.118)	Yes for the top 3; ranks 4–5 are noise (≈0.04).
3. Free text	"the supplier sent parts late and the line may stop, what should I do"	DOC036 (0.166), DOC038 (0.144), DOC013 (0.099)	Yes. Returns the premium freight procedure.

Table 1. Summary of the three tests. Full Top-5 and console captures are in the /evidencias/ folder.

The recommendations are coherent because related documents share specific terms such as *andon*, *MES terminal* or *supplier*. They also cross categories, linking a policy with its

training and real incidents, which is exactly what a person needs to act.

2. Areas of Opportunity

- **Synonyms and meaning:** TF-IDF matches words, not ideas. In test 3, the conveyor line stop appeared only because it shares "line stop". Sentence embeddings (Path B) or stemming would capture meaning.
- **Low-score noise:** in test 2, ranks 4–5 scored about 0.04. A minimum similarity threshold would show fewer but better results.
- **Short documents and small corpus:** with 40 short texts, few words overlap and scores stay low (0.05–0.29). A real repository would give more stable vectors.
- **Corpus bias:** Supply Chain has only 4 documents, so it is recommended less often. Categories should be balanced as the repository grows.
- **Real-world environment:** add metadata (plant, line, date, version), role-based filters and access control so confidential documents are only shown to authorized roles, plus continuous evaluation with user feedback (useful / not useful) and precision@5.

3. Organizational Reflection Video

The video explains why this type of recommendation reduces rework and speeds up decisions, the limitations observed, and the improvements for a real environment.

> **Video link:** https://youtu.be/f-_jTOZWBf8